Unsupervised Clustering of Alzheimer's/Dementia Patients

from Brain MRI and Clinical Data: A Full Pipeline Report

OASIS-1 cross-sectional dataset ($n = 416$ patients, $n = 216$ complete-case)

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November 10, 2026

Abstract

This report documents an end-to-end investigation of whether brain-MRI-derived image features, alone or combined with clinical assessments, can recover clinically meaningful patient subgroups, specifically groups that align with the Clinical Dementia Rating (CDR) for the OASIS-1 cross-sectional cohort. Five MRI planes per patient are reduced to per-image PCA components and combined using a family of robust, mixed-type distance metrics (Robust Mahalanobis, Generalised Gower, Related Metric Scaling) and a fast k-medoids clustering engine. Nine experimental stages are reported: single-image-type clustering, merged 5-image clustering, clinical+imaging fusion under Generalised Gower, two-image concatenation, Related Metric Scaling (RelMS), an image-redundancy correlation analysis, a leave-one-image-out sensitivity sweep, a full cluster-profiling analysis of the flagship 5-image+clinical configuration, and a per-block distance-sensitivity study exploiting a package update from the advisor. Across every configuration tested, the recovered clusters show clear, substantial structure: cluster membership is strongly and consistently associated with sex (Cramér's V up to 0.66) and total intracranial volume (eTIV/ASF, η^2 up to 0.56), both large effects by conventional standards, and with education and SES in several configurations. That structure does not, however, align with the Clinical Dementia Rating (CDR): Adjusted Rand Index (ARI) against CDR stays within roughly $[-0.02, +0.06]$ and Normalised Mutual Information (NMI) within $[0.01, 0.10]$ throughout, close to what a random partition would achieve on that axis specifically. Two methodological anomalies encountered along the way are documented and flagged rather than resolved. Taken together, these results do not point to an absence of structure in the data. The clustering method works, and works consistently: it reliably finds real patient groups, just groups organised around demographic and anatomical variation rather than around dementia severity.

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1 Introduction

Digital/computational pipelines that turn raw, high-dimensional signals (text, images) into a small set of validated or engineered features, and then cluster those features with a distance metric suited to mixed data types, are increasingly used across domains.

This thesis applies this approach to a high-dimensional input of its own: brain MRI images. The scientific question is whether image-derived structural features of the brain, alone or combined with standard clinical assessments, can recover clinically meaningful patient subgroups for Alzheimer's disease, in particular, subgroups that line up with the Clinical Dementia Rating (CDR), a standard severity scale. This thesis finds that clustering consistently succeeds at recovering *some* axis of real patient structure, just not the CDR axis. Under every combination of image features, clinical features, and distance metric tried, the clusters produced are large, stable, and strongly differentiated by sex and brain volume (and, in several configurations, by education and socioeconomic status), but fail to line up with disease severity. This report documents the full pipeline, methodology, and every experiment run, in order, together with the figures and numeric results each stage produced, with particular emphasis on *which* features actually drive each clustering, not only on how well each clustering matches CDR.

2 Data and Preprocessing

2.1 OASIS-1 Dataset

The Open Access Series of Imaging Studies (OASIS-1) is a cross-sectional collection of 416 subjects aged 18–96, including cognitively normal individuals and patients with early Alzheimer's disease. The Clinical Dementia Rating (CDR) is used as the post-hoc evaluation label: CDR = 0 (healthy), 0.5 (very mild dementia), 1 (mild dementia), 2 (moderate dementia). CDR is known for only 235 of the 416 subjects and is **never** used as a clustering input; it is used only to evaluate the clusters after the fact, via Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI).

2.2 Five MRI Image Types

Each patient has up to five MRI planes:

- `cor`: coronal
- `gfc_sag`: sagittal, atlas-registered (Talairach space)
- `gfc_tra`: transverse/axial, atlas-registered
- `masked_tra`: transverse, brain-masked (non-brain tissue removed)
- `sbj_sag`: native (subject-space) sagittal, *not* atlas-registered, retains individual anatomical variation that registration would otherwise remove

2.3 HOG Feature Extraction

Each image is padded to a square aspect ratio (centred, zero-padded on the shorter dimension), resized to 128×128 pixels, and converted to Histogram-of-Oriented-Gradients (HOG) features: 9 orientations, 8×8 px/cell, 2×2 cells/block $\Rightarrow$ **8,100 features per image**, or **40,500 features per patient** across all five image types.

2.4 Per-Image PCA

HOG features are reduced per image type with a two-pass PCA strategy: pass 1 finds, independently for each image type, the minimum number of components needed to reach 59.2% explained variance; the maximum of those minima across the five types (110) is taken as a unified target; pass 2 refits every image type to exactly 110 components, so all five types are directly comparable dimensionally.

The five resulting (416×110) PCA matrices are the base feature blocks for every downstream experiment.

2.5 Clinical Variables

Eight clinical variables accompany the imaging features: Age, M/F, Education, Socioeconomic Status (SES), MMSE (Mini-Mental State Exam), eTIV (estimated total intracranial volume), nWBV (normalised whole-brain volume), and ASF (atlas scaling factor). Patients missing Education, SES, or MMSE are dropped (no imputation) for any experiment that uses clinical data, leaving **216 complete-case patients**, all of whom have a known CDR.

3 Clustering Methodology

All clustering uses a family of robust, mixed-type distance metrics and a fast k-medoids engine (`robust_mixed_dist` / `db_robust_clust`, developed by the thesis advisor, Fabio Scielzo-Ortiz).

- **Robust Mahalanobis** distance (trimmed, $\alpha = 0.05$) for a single quantitative block, e.g. one image type's 110 PCA components.
- **Generalised Gower (G-Gower)** distance for mixing quantitative, binary, and categorical blocks (e.g. PCA components + clinical variables) into one distance matrix.
- **Related Metric Scaling (RelMS)**, introduced partway through this project specifically to address a block-dominance problem found with G-Gower (Section 4.3): RelMS treats each declared block (e.g. each image type, and the clinical quantitative block) *separately*, rescales each by its own geometric variability, and only then combines them, so no single block dominates purely because it has more columns.
- **SampleDistClustering / FasterPAM k-medoids**: a fast Partitioning-Around-Medoids implementation that subsamples for tractability on larger matrices and returns real patients as cluster medoids rather than synthetic centroids.

Evaluation. Every experiment is evaluated the same way: silhouette score (geometric separation, computed on all clustered patients), and Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) against CDR (computed only on the subset with known CDR). CDR is multiplied by 2 to obtain integer categories for ARI/NMI. A "spread" statistic (the range of % CDR-healthy across clusters) is also tracked as a secondary, more interpretable indicator of separation. $k \in \{2, 3, 4\}$ is tested throughout, occasionally up to $k = 8$ for formal k-selection diagnostics.

4 Experiments and Results

Nine experimental stages are reported below, in the order they were run. Two further branches were also explored but are omitted from this report as exploratory/secondary work: a Tucker tensor-decomposition alternative to per-image PCA, and a two-stage PCA + Robust Mahalanobis dimensionality-planning exercise.

4.1 Single-Image-Type Clustering

Each of the five image types is clustered on its own 110-component PCA block, with both Euclidean and Robust Mahalanobis distance, at $k = 3$ and $k = 4$ (20 combinations total).

Table 1: Single-image-type clustering results (20 combinations). Best result overall (highlighted in the text) is `masked_tra`, Euclidean, $k = 3$.

Image type	Distance	k	Silhouette	ARI	NMI	Spread (%)
Coronal	Euclidean	3	0.0223	-0.0079	0.0221	18.4
Coronal	Euclidean	4	0.0079	-0.0244	0.0368	31.2
Coronal	Robust Mahalanobis	3	0.0005	-0.0135	0.0127	10.0
Coronal	Robust Mahalanobis	4	0.0011	0.0094	0.0289	29.1
Sagittal (atlas)	Euclidean	3	0.0235	0.0170	0.0189	18.8
Sagittal (atlas)	Euclidean	4	0.0210	0.0009	0.0313	19.2
Sagittal (atlas)	Robust Mahalanobis	3	0.0029	-0.0051	0.0070	0.9
Sagittal (atlas)	Robust Mahalanobis	4	0.0023	-0.0014	0.0158	8.0
Transverse (atlas)	Euclidean	3	0.0009	-0.0121	0.0442	33.0
Transverse (atlas)	Euclidean	4	0.0104	-0.0031	0.0485	34.7
Transverse (atlas)	Robust Mahalanobis	3	0.0014	-0.0032	0.0124	4.8
Transverse (atlas)	Robust Mahalanobis	4	0.0005	-0.0044	0.0097	5.3
Masked transverse	Euclidean	3	0.0257	0.0597	0.0851	42.8
Masked transverse	Euclidean	4	0.0062	0.0279	0.0814	47.1
Masked transverse	Robust Mahalanobis	3	0.0004	-0.0013	0.0140	10.9
Masked transverse	Robust Mahalanobis	4	-0.0001	0.0020	0.0162	14.9
Sagittal (native)	Euclidean	3	0.1348	-0.0050	0.0164	12.3
Sagittal (native)	Euclidean	4	0.1112	-0.0041	0.0140	13.0
Sagittal (native)	Robust Mahalanobis	3	0.0031	0.0060	0.0122	8.7
Sagittal (native)	Robust Mahalanobis	4	-0.0005	0.0116	0.0187	13.9

The best CDR alignment found anywhere in this entire project comes from this stage: `masked_tra` alone, Euclidean distance, $k = 3$ (ARI = 0.060, NMI = 0.085). Robust Mahalanobis performs consistently worse than Euclidean at this stage: most of its ARI values sit near zero or negative, and silhouette scores collapse to near zero, indicating the robust covariance estimate is not finding useful structure in a single 110-dimensional PCA block.

4.2 Merged 5-Image Clustering

All five image types are concatenated into a single 416×550 feature matrix and clustered with Euclidean and Robust Mahalanobis distance.

Table 2: Merged 5-image clustering (all 550 PCA features, equal weight per column).

Distance	k	Silhouette	ARI	NMI	Spread (%)
Euclidean	3	0.0371	-0.0072	0.0220	9.8
Euclidean	4	0.0366	0.0129	0.0498	33.0
Robust Mahalanobis	3	0.0014	-0.0006	0.0284	23.4
Robust Mahalanobis	4	-0.0011	-0.0002	0.0342	34.5

Simply merging all five image types performs *worse* than the best single image type alone (Section 4.1), an early sign that naively concatenating image blocks dilutes rather than reinforces the signal, motivating the later RelMS approach.

4.3 Clinical + Imaging Fusion via Generalised Gower

All-5-images-merged PCA features, each single image type's PCA features, and clinical-only features are each combined with the 8 clinical variables via Generalised Gower distance (`SampleDistClustering`, `metric='ggower'`).

Table 3: Generalised Gower clustering: condensed view. **Every** image-type variant (all-5-merged, each single image type, and even the clinical-only baseline) produced numerically identical results down to the cluster sizes; three representative rows are shown.

Feature set	k	Silhouette	ARI	NMI
5 images + clinical	3	0.3272	0.0160	0.0216
5 images + clinical	4	0.3014	0.0006	0.0156
Coronal + clinical (representative)	3	0.3284	0.0160	0.0216
Coronal + clinical (representative)	4	0.3034	0.0006	0.0156
Clinical only (no imaging)	3	0.3286	0.0160	0.0216
Clinical only (no imaging)	4	0.3039	0.0006	0.0156

Anomaly. This is the first of two methodological anomalies flagged in this project (see Section 5). Every image+clinical variant under Generalised Gower, whether it includes 550, 110, or 0 image PCA columns, produces the *exact same* silhouette, ARI, NMI, and cluster assignments as the clinical-only baseline. This strongly suggests the clinical block was dominating the Generalised Gower distance so completely under this particular block configuration that the image PCA columns contributed essentially zero discriminative signal, most likely a scaling/normalisation interaction between a 110–550-dimensional quantitative image block and a 5-dimensional quantitative clinical block within the same distance subspace. This observation directly motivated switching to RelMS (Section 4.5), which rescales each declared block independently before combining.

4.4 Two-Image Concatenation

As an intermediate step, two image types most likely to be complementary (`masked_tra` + `cor`) were simply concatenated (Euclidean distance, no explicit block weighting).

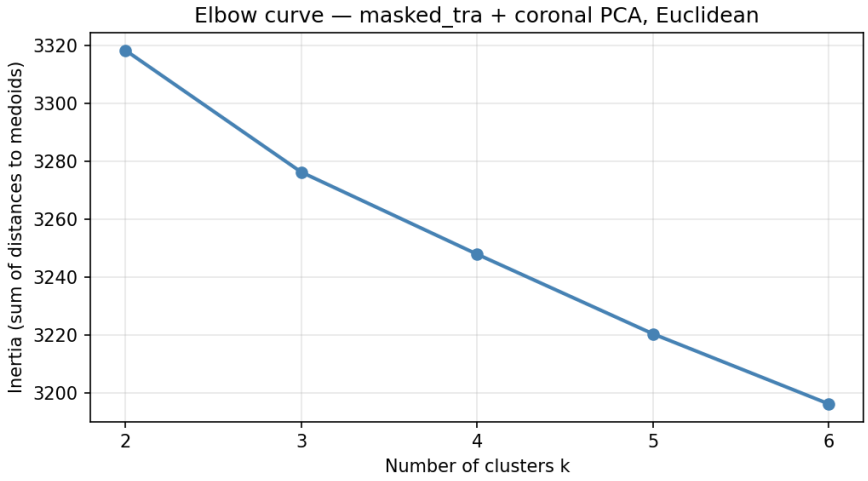


Figure 1: Elbow curve (FasterPAM loss vs. k) for the `masked_tra+cor` concatenated feature matrix, Euclidean distance.

Results stayed near zero (ARI in $[-0.03, 0.01]$ across $k = 3, 4$), confirming that naive concatenation, without RelMS-style block rescaling, does not help even for a carefully chosen pair of image types.

4.5 Related Metric Scaling (RelMS)

RelMS treats each image type (and the clinical quantitative variables) as a separate block, rescaling each by its own geometric variability before combining, specifically to correct the block-dominance problem found in Section 4.3. This was tested on: all 5 images with no clinical data; all 5 images +

clinical (run twice, once with an initial M/F binary/Sokal encoding, and again after a correction that moved M/F into the multiclass/Hamming block alongside Education and SES, which is the methodologically correct treatment for a nominal two-category variable); and the two most promising image pairs, with and without clinical data.

Table 4: RelMS results: all-5-image blends, with and without clinical data, before and after the M/F encoding correction.

Configuration	k	**Silhouette**	**ARI**	**NMI**
5 images, no clinical	3	0.0183	-0.0103	0.0233
5 images, no clinical	4	0.0106	-0.0056	0.0240
5 images + clinical (initial M/F encoding, $n = 216$)	3	0.0159	0.0092	0.0206
5 images + clinical (initial M/F encoding, $n = 216$)	4	0.0112	0.0202	0.0268
5 images + clinical (corrected M/F encoding, $n = 216$)	2	0.0262	0.0283	0.0107
5 images + clinical (corrected M/F encoding, $n = 216$)	3	0.0091	0.0176	0.0188

Table 5: RelMS results: best two-image combinations, with and without clinical data.

Combination	**+Clinical**	k	n	**Silhouette**	**ARI**	**NMI**
masked_tra+cor	No	3	416	0.0068	-0.0028	0.0444
masked_tra+cor	No	4	416	0.0086	0.0141	0.0437
masked_tra+gfc_tra	No	3	416	0.0065	-0.0287	0.1009
masked_tra+gfc_tra	No	4	416	0.0057	-0.0359	0.0624
masked_tra+cor+clinical	Yes	3	216	0.0200	0.0323	0.0422
masked_tra+cor+clinical	Yes	4	216	0.0138	0.0409	0.0414
masked_tra+gfc_tra+clinical	Yes	3	216	0.0203	0.0157	0.0201
masked_tra+gfc_tra+clinical	Yes	4	216	0.0138	0.0220	0.0411

RelMS moves the numbers slightly upward relative to plain Generalised Gower and simple concatenation: the best result here, `masked_tra+cor+clinical` at $k = 4$, reaches ARI = 0.041, NMI = 0.041, but still remains far from a meaningful CDR-aligned partition. RelMS fixed the block-dominance *artifact*, but did not fix the underlying *absence* of CDR-relevant signal in the image features themselves.

4.6 Image-Type Redundancy Analysis

To understand which image types carry unique versus overlapping information, the pairwise-patient Euclidean distance matrix of each image type (upper triangle, 86,320 unique patient pairs) was correlated against every other image type's distance matrix.

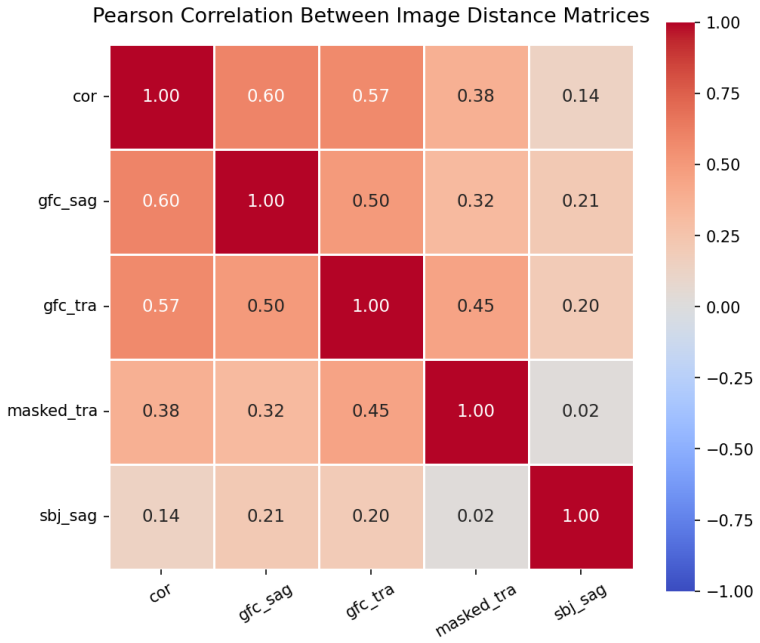


Figure 2: Pearson correlation between the pairwise-patient-distance matrices of the five image types. Higher values mean two image types rank patient similarity almost the same way (redundant); lower values mean they capture different structural information (complementary).

Table 6: Image-type pairwise distance-matrix correlations, ranked highest to lowest.

Image-type pair	**Pearson r**
Coronal ↔ Sagittal (atlas)	0.5981
Coronal ↔ Transverse (atlas)	0.5735
Sagittal (atlas) ↔ Transverse (atlas)	0.4957
Transverse (atlas) ↔ Masked transverse	0.4474
Coronal ↔ Masked transverse	0.3808
Sagittal (atlas) ↔ Masked transverse	0.3158
Sagittal (atlas) ↔ Sagittal (native)	0.2132
Transverse (atlas) ↔ Sagittal (native)	0.1974
Coronal ↔ Sagittal (native)	0.1374
Masked transverse ↔ Sagittal (native)	0.0161

The most redundant pair is `cor` ↔ `gfc_sag` ($r = 0.598$); the most complementary/independent pair is `masked_tra` ↔ `sbj_sag` ($r = 0.016$), meaning the native (unregistered) sagittal images carry the most anatomically unique signal, distinct from every other type. This finding directly motivated which image pairs were prioritised in Section 4.5 and which single blocks were targeted in the per-block distance sensitivity study (Section 4.9).

4.7 Leave-One-Image-Out Sweep

Building on the redundancy finding above, each image type was removed in turn from the RelMS 5-image blend to see whether it was contributing real CDR-relevant signal or mostly noise.

Table 7: Leave-one-image-out: mean change in ARI/NMI (vs. the full 5-image RelMS baseline) across $k = 3, 4$, when the named image type is removed. Positive = removing that image *improved* CDR alignment (it was diluting signal); negative = removing it *hurt* alignment (it was contributing real signal).

Image type removed	**Mean ΔARI**	**Mean ΔNMI**
Masked transverse	+0.0046	+0.0050
Coronal	+0.0009	-0.0128
Sagittal (native)	+0.0003	+0.0046
Sagittal (atlas)	-0.0050	-0.0011
Transverse (atlas)	-0.0074	-0.0010

Removing `masked_tra` or `sbj_sag` tends to help slightly, while removing `gfc_sag` or `gfc_tra` tends to hurt, consistent with the redundancy finding (`gfc_sag`/`gfc_tra` correlate strongly with `cor` and contribute mostly overlapping signal, whereas `masked_tra`/`sbj_sag` are more independent but also noisier for this task). All deltas, however, are small ($|\Delta\text{ARI}| \leq 0.007$), well within the noise floor observed throughout this project.

4.8 Flagship Experiment: Full Cluster Profiling (5 Images + Clinical, RelMS)

The corrected RelMS 5-image+clinical configuration ($n = 216$, 558 features: 5×110 image PCA + 5 clinical quantitative + 3 clinical categorical) was profiled in the most depth: cluster-by-cluster clinical/demographic breakdowns, effect-size rankings of which features actually separate the clusters, and (secondarily) optimal-k selection and geometric separation diagnostics. The emphasis here is deliberately on *what the clusters are made of*, which features differ between them and by how much, rather than on clustering-quality scores alone.

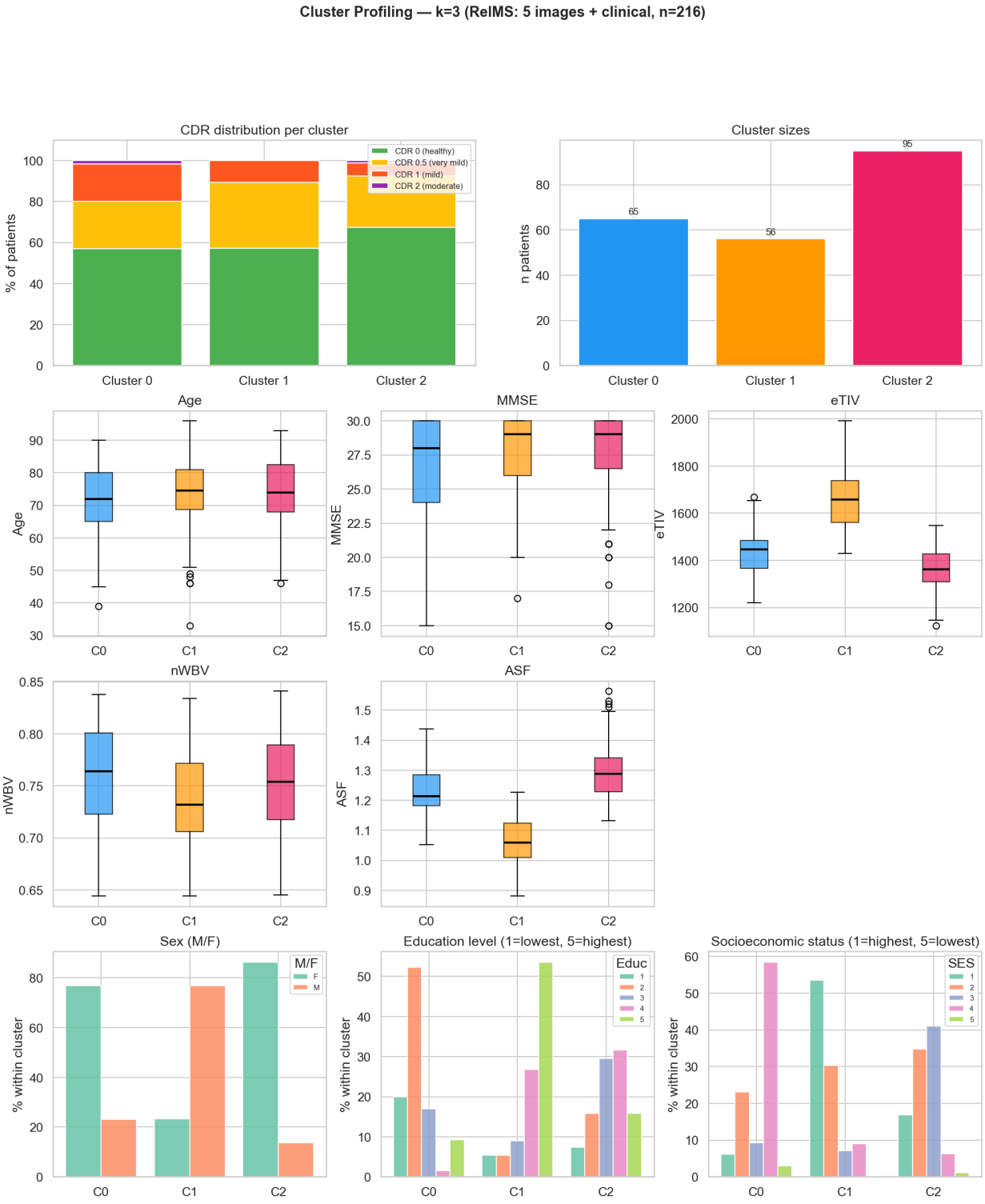


Figure 3: Full cluster profile at $k = 3$: CDR distribution and cluster sizes (top row), boxplots of Age, MMSE, eTIV, nWBV and ASF (middle rows), and categorical breakdowns of sex, education and SES (bottom row). Sex and eTIV visibly separate the three clusters; CDR, Age and MMSE do not.

Table 8: Cluster profiles at $k = 3$, 5 images + clinical, RelMS, $n = 216$.

Cluster	n	CDR0	CDR0.5	CDR1	CDR2	Age	MMSE	eTIV	Educ	SES	%F
C0	65	56.9	23.1	18.5	1.5	71.3	26.75	1430	2.28	3.29	76.9
C1	56	57.1	32.1	10.7	0.0	72.4	27.62	1661	4.18	1.71	23.2
C2	95	67.4	25.3	6.3	1.1	73.2	27.54	1359	3.33	2.40	86.3

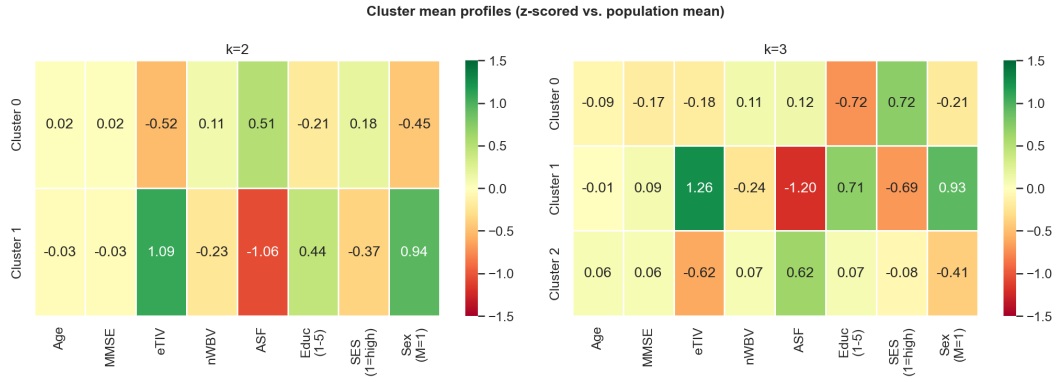


Figure 4: Cluster mean profile (z-scored vs. population mean), $k = 2$ and $k = 3$. Cells further from zero indicate a feature on which that cluster deviates most from the overall population. Sex (M/F) and eTIV are consistently the strongest differentiators.

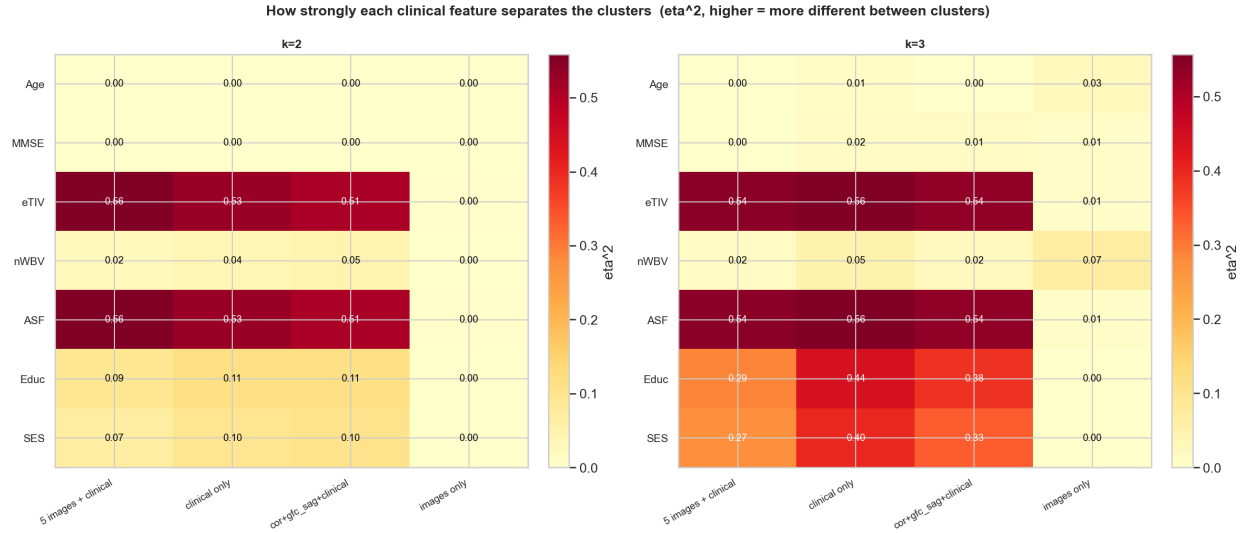


Figure 5: Effect size (η^2, higher = stronger separation) of each clinical feature across four feature-set configurations (5 images+clinical, clinical-only, best 2-image+clinical, images-only). eTIV and sex consistently show the largest effect sizes.

Which features actually drive the clusters. At $k = 3$, sex shows a large effect (Cramér's $V = 0.560$) and eTIV/ASF a large effect ($\eta^2 = 0.543/0.542$); education and SES are also large ($\eta^2 = 0.287$ and 0.273). Age and MMSE are negligible ($\eta^2 \approx 0.000$), and CDR itself (external, never used in clustering) registers only $V = 0.135$. The same ordering holds at $k = 2$ (sex $V = 0.642$, eTIV/ASF $\eta^2 = 0.558$) and across the other three configurations in Figure 5. In other words, the clustering is not failing to find structure: it is finding real, large, and highly reproducible structure, organised around sex, intracranial volume, and socioeconomic variables, rather than around dementia severity.

Geometric diagnostics (secondary). Optimal-k selection (elbow, silhouette, Davies-Bouldin, $k = 2$ to 8) agrees on $k = 2$ as statistically optimal, with $k = 3$ also reported for its clinical interpretability (healthy / very mild / mild–moderate); a 2D PCA scatter of the 558-column feature matrix shows PC1 alone explains ∼99% of variance, so cluster boundaries fall along a single dominant axis rather than around visually distinct groupings in 2D. These geometric scores are reported for completeness but are secondary to the feature-level findings above.

Anomaly. A second methodological anomaly, found in a related Tucker-decomposition configuration during this same profiling phase, is worth flagging: Robust Mahalanobis at $k = 3, 4$ occasionally

produced degenerate singleton-outlier clusters (sizes such as $\{1, 414, 1\}$) while reporting a deceptively high silhouette score (~ 0.78). This is an artifact of the metric isolating one or two extreme outliers into their own "cluster," not evidence of genuine structure, and a useful reminder that silhouette alone can be misleading whenever a partition is highly imbalanced.

4.9 Per-Block Distance Update (Package 0.1.27)

Partway through this project, the advisor released an update to `robust-mixed-dist` (0.1.27) that allows *different* distances for different quantitative blocks within RelMS, rather than one distance string applied to every block. All four notebook 18 configurations (5 images + clinical, clinical only, images only, and the best 2-image combination + clinical) were rebuilt under the updated package, applying a single, consistent rule wherever a clinical quantitative block exists: the image blocks remain on Euclidean distance exactly as before, and the clinical block switches to Robust Mahalanobis distance, the professor's own example of the new capability. The images-only configuration has no clinical block, so it is numerically unchanged from notebook 18.

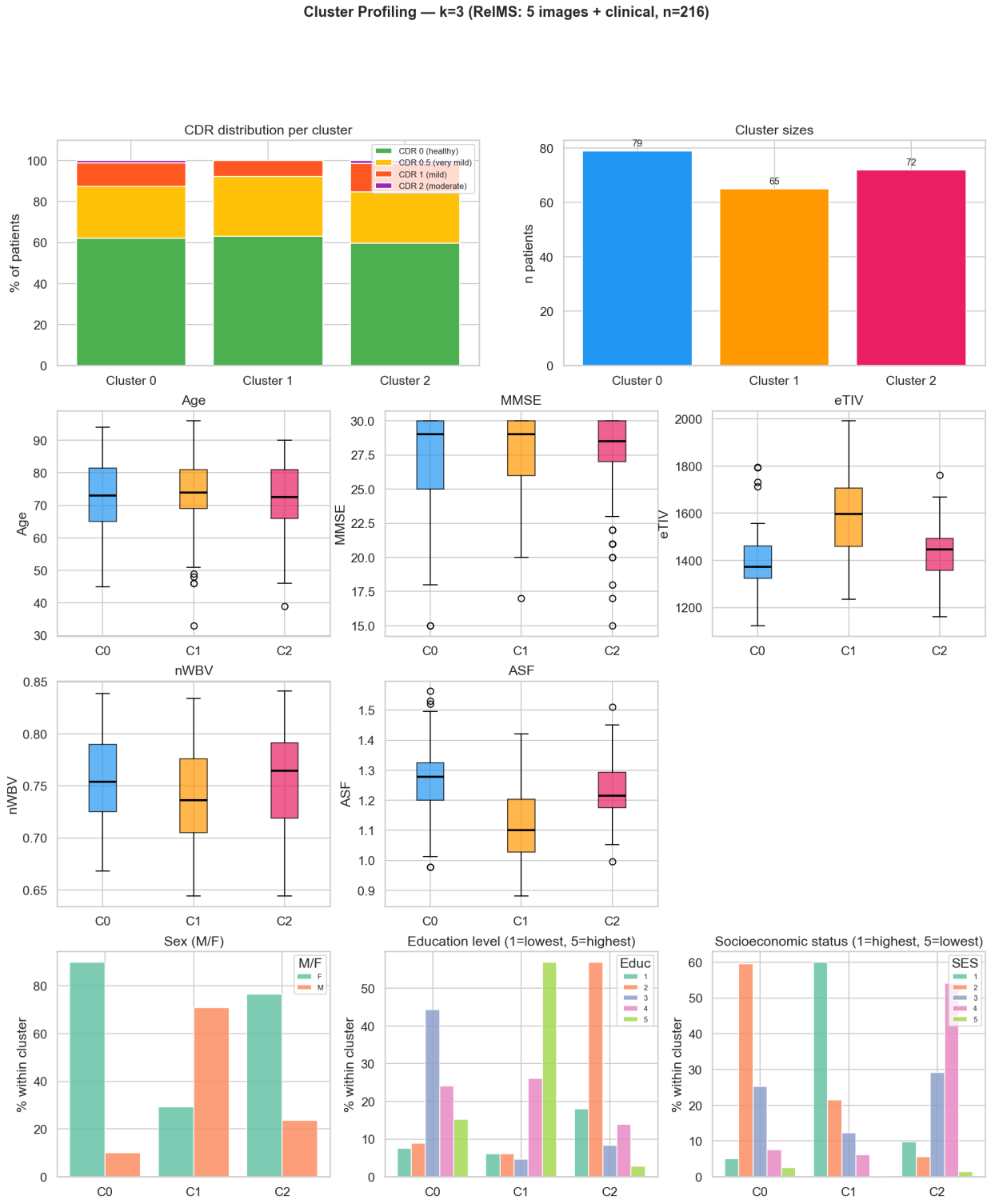


Figure 6: Full cluster profile at $k = 3$ under the updated (Robust-Mahalanobis-clinical-block) 5-image+clinical configuration (the direct counterpart to Figure 3). The same pattern holds: sex and eTIV visibly separate the clusters, CDR does not.

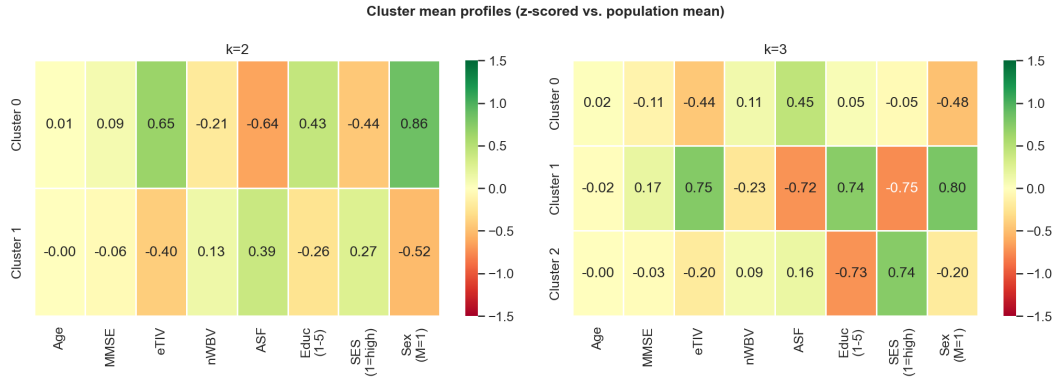


Figure 7: Cluster mean profile (z-scored), $k = 2$ and $k = 3$, under the updated configuration (the direct counterpart to Figure 4).

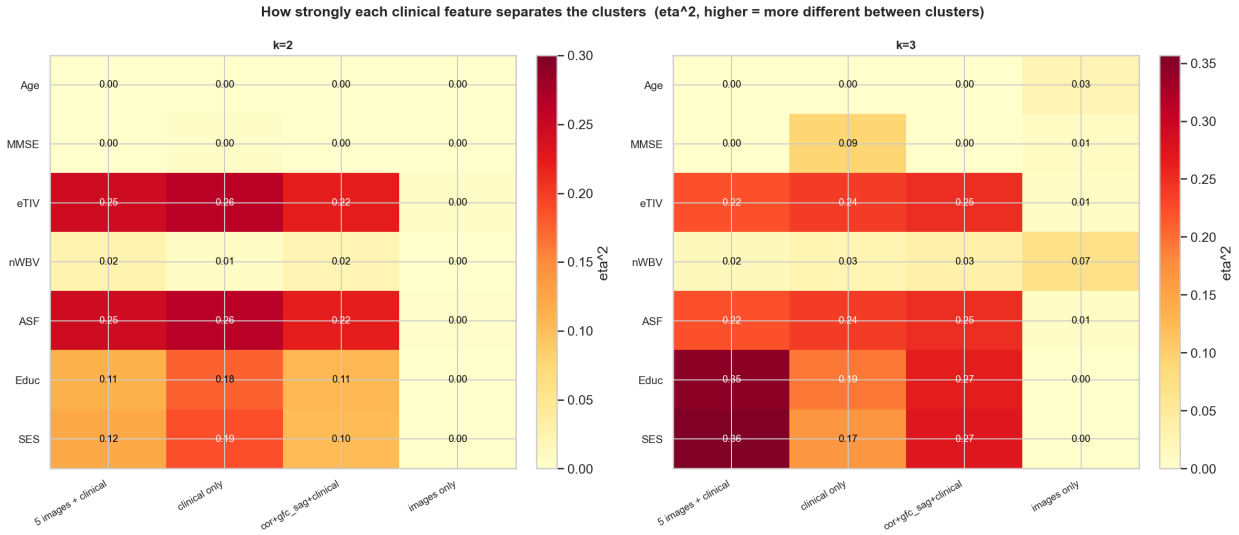


Figure 8: Effect size (η^2) of each clinical feature across the four updated (Robust-Mahalanobis-clinical-block) configurations (the direct counterpart to Figure 5). The same features (sex, eTIV) dominate as under the original Euclidean setup.

Feature separation is preserved under the update. The same large, reproducible effects seen in Section 4.8 persist here: sex and eTIV/ASF remain the dominant differentiators in every one of the four updated configurations, at essentially the same magnitude as under plain Euclidean (compare Figure 8 to Figure 5). Switching the clinical block's distance function changes *which patients* fall into each cluster (see the geometric comparison below), but not *what kind* of structure the clustering finds overall.

Geometric diagnostics (secondary). Switching the clinical block to Robust Mahalanobis made three of the four configurations slightly worse by silhouette/ARI against CDR and left the fourth (images-only) unchanged; only clinical-only at $k = 3$ improved, and by an amount (ARI $0.018 \rightarrow 0.038$) too small to read as a real effect.

Table 9: All four notebook 18 configurations, before (Euclidean throughout) and after (clinical block switched to Robust Mahalanobis) the package update. Images-only is identical in both columns by construction, since it has no clinical block. Reported for completeness; see the feature-level discussion above for the more informative comparison.

Configuration	k	Sil. (old)	Sil. (new)	ARI (old)	ARI (new)
5 images + clinical	2	0.0262	0.0098	0.0283	0.0043
5 images + clinical	3	0.0091	0.0053	0.0176	-0.0063
Clinical only	2	0.0292	0.0118	0.0245	-0.0056
Clinical only	3	0.0130	0.0064	0.0177	0.0384
Images only (unchanged)	2	0.0510	0.0510	-0.0040	-0.0040
Images only (unchanged)	3	0.0376	0.0376	-0.0205	-0.0205
Best combo + clinical	2	0.0269	0.0109	0.0320	0.0058
Best combo + clinical	3	0.0112	0.0065	0.0161	-0.0045

Diagnosing the small CDR movements above required looking at the clinical variables themselves: `eTIV` and `ASF` are correlated at $r = -0.990$ in this cohort (ASF is computed as an atlas scaling factor derived from head size, so the two are near-mathematical duplicates), making the clinical block's 5×5 correlation matrix nearly singular (smallest eigenvalue 0.010 vs. largest 2.262, condition number ≈ 218). Robust Mahalanobis distance inverts that correlation structure, which amplifies estimation noise along the near-degenerate `eTIV/ASF` direction. Euclidean distance, which never inverts anything, is not exposed to this instability. A targeted follow-up check on the clinical-only block confirmed the mechanism is real rather than arbitrary: Robust Mahalanobis gives MMSE (the single clinical variable most directly tied to dementia severity) roughly 5.5× more influence over the resulting clusters than Euclidean does ($\eta^2 = 0.093$ vs. 0.017 at $k = 3$), which is exactly why clinical-only improved slightly at that k. Even so, this extra influence is still far too small, on its own, to produce a meaningfully CDR-aligned partition. **The distance metric is not the bottleneck**: it can be tuned to give more weight to clinically relevant variables, and when it does, the effect on CDR alignment is still marginal, confirming that the limiting factor is the absence of CDR-discriminative signal in the feature set itself rather than any particular choice of distance.

5 Discussion

5.1 Central Finding

Across all nine stages (single image types, merged images, Generalised Gower, RelMS, two-image combinations, a leave-one-image-out sweep, and the post-update per-block distance comparison across all four notebook 18 configurations), **the clustering consistently recovers real, large, reproducible structure, but that structure is organised around sex and total intracranial volume (eTIV/ASF), not around CDR**. The effect sizes behind this are substantial by conventional standards: sex reaches Cramér's V up to 0.66 and eTIV/ASF reach η^2 up to 0.56 across every configuration tested (Figures 5 and 8), with education and SES also registering large effects (η^2 up to 0.44) at several configurations. Age and MMSE, by contrast, are consistently negligible ($\eta^2 \approx 0.00$–0.02). Measured specifically against CDR, this same structure is weak: ARI against CDR stays roughly within $[-0.02, +0.06]$ and NMI within $[0.01, 0.10]$ throughout, i.e. barely above what a random partition would achieve on that axis, and the best single CDR-aligned result across the entire project is `masked_tra` alone, Euclidean distance, $k = 3$ (ARI = 0.060, NMI = 0.085). The picture that emerges is best described as a *mismatch of axes* rather than an absence of structure: HOG+PCA features recovered from these MRI images carry a strong, stable signal about demographic and anatomical variation, and that signal survives every distance metric, block weighting, and image combination tried in this project. It simply is not the signal clinicians use to grade dementia severity.

5.2 Two Flagged Anomalies

1. **Generalised Gower block dominance** (Section 4.3): every image+clinical variant under Generalised Gower reproduced the clinical-only baseline exactly, suggesting the clinical block dominated that particular distance formulation so completely that image features contributed no signal. This was the direct motivation for adopting RelMS, and remains worth a closer look at the underlying scaling interaction within the Generalised Gower implementation.
2. **Degenerate outlier clusters under Robust Mahalanobis** (Section 4.8): in a related Tucker-decomposition configuration, Robust Mahalanobis occasionally isolated 1–2 extreme outliers into their own cluster while reporting an artificially high silhouette score, a caution against trusting silhouette alone without checking cluster-size balance.

6 Conclusion

This project built a full pipeline (HOG feature extraction, per-image PCA, and a systematic sweep across Robust Mahalanobis, Generalised Gower, and RelMS distance metrics, single- and multi-image feature combinations, and, most recently, per-block distance-function sensitivity) to test whether brain MRI structure, with or without standard clinical variables, can recover Alzheimer's disease severity groups in an unsupervised setting. The clustering itself works well: every configuration tested recovers clusters with large, consistent, and reproducible separation on sex (Cramér's V up to 0.66) and total intracranial volume (eTIV/ASF, η^2 up to 0.56), and several configurations also separate cleanly on education and SES. What those clusters do not do, consistently, is line up with CDR. The strongest single CDR-aligned result (`masked_tra`, Euclidean, $k = 3$, ARI $= 0.060$) and the redundancy/leave-one-out analyses point toward `masked_tra` and `sbj_sag` as the most anatomically distinctive image types, while the effect-size analysis points toward sex and intracranial volume, rather than CDR-adjacent variables, as the strongest drivers of the structure actually recovered. Two methodological anomalies were documented rather than silently resolved, for future investigation. Overall, this is a well-controlled result that narrows the problem rather than leaving it open. The bottleneck is not the choice of distance metric, block weighting, or image combination (all of those reliably find real structure), but the absence, in this particular image representation, of a signal that happens to align with dementia severity specifically.